Questionnaire on VQFlow

for Developers and Maintainers (Writing or Modifying SELECT Statements)

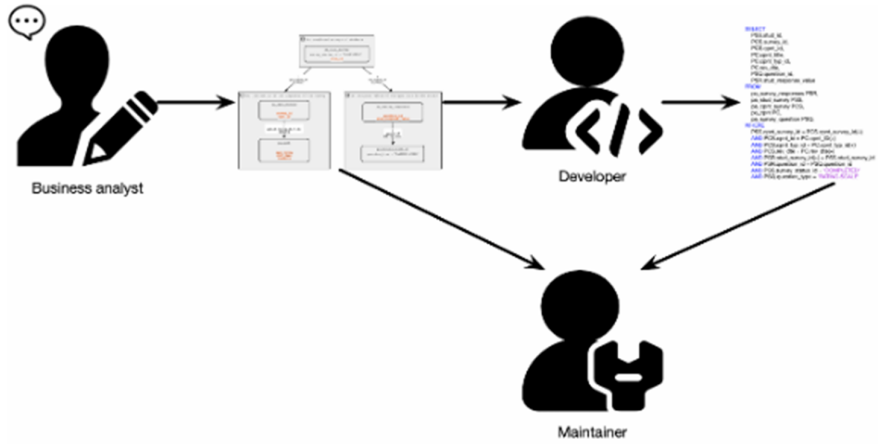

Table reports generation requires requirements engineers describing the query requirements, i.e., how to retrieve the required data from the database. Traditionally, tables are used to describe the fetch logic for each column. However, when SQL queries become complex, for example, more tables and more filtering conditions are involved, the relationships between tables and columns become intricate, which makes describing query requirements extremely difficult.

To solve such a problem, we developed a tool called VQFlow (**V**isual **Q**uery **Flow**) to help requirements engineers describe query requirements quickly and clearly. Before using VQFlow, requirements engineers should know the structure of the tables involved and the relationships between them. VQFlow represents requirements as a type of data flow diagram. Based on this diagram, developers can quickly and correctly understand query requirements and write corresponding SQL SELECT statements. Maintainers can also use this diagram to quickly understand the business logic behind the SELECT statements and easily modify them. VQFlow is expected to significantly reduce the cost of communication between requirements engineers, developers and maintainers.

In order to investigate the effectiveness of VQFlow, we invite you to answer this questionnaire. Before it, you need to read the three cases provided by us: each case corresponds to a query requirement which includes a one-sentence summary of the requirement, the tabular description of the query requirement (the traditional method), the VQFlow description of the query requirement, the files that can be imported into VQFlow, and the corresponding SELECT Statement. Three cases: https://github.com/vqflow/questionnaire/blob/main/Three_cases.zip

In addition, more information about VQFlow is as following.

VQFlow tool: https://vqflow.github.io/

VQFlow instructions: https://github.com/vqflow/questionnaire/blob/main/Instruction_manual.pdf

Demo for VQFlow: https://github.com/vqflow/questionnaire/blob/main/demo.mp4

Thank you for taking time to answer this questionnaire!

Read three cases (including two forms of requirement descriptions) and answer the following questions.

1. (Multiple choice) When understanding query requirements, what problems do you see with the query requirements described in the traditional method (using tables)?

☐ It takes longer to understand the requirements.

☐ When requirements change, it takes longer to re-understand the requirements.

☐ When requirements change, the modified SQL code is prone to error due to insufficient understanding of requirements.

☐ Frequent communication with requirements engineers is required to understand requirements.

☐ It is difficult to clearly understand the relationships between tables.

☐ It is difficult to clearly understand the relationships between columns.

☐ Others: _____.

2. (Single option) How do you rate VQFlow affecting the efficiency (length of time) of understanding query requirements when writing SQL SELECT statements compared to the traditional method (using tables)?

○ Great increment.

○ Small increment.

○ No change.

○ Small decrement.

○ Great decrement.

3. (Single option) How do you rate VQFlow affecting the difficulty of understanding query requirements when writing SQL SELECT statements compared to the traditional method (using tables)?

○ Great decrement.

○ Small decrement.

○ No change.

○ Small increment.

○ Great increment.

4. (Single option) How do you rate VQFlow affecting the efficiency (length of time) of understanding query requirements when maintaining other people's SQL SELECT statements compared to the traditional method (using tables)?

○ Great increment.

○ Small increment.

○ No change.

○ Small decrement.

○ Great decrement.

5. (Single option) How do you rate VQFlow affecting the difficulty of understanding requirements when maintaining other people's SQL SELECT statements compared to the traditional method (using tables)?

○ Great decrement.

○ Small decrement.

○ No change.

○ Small increment.

○ Great increment.

6. (Multiple choice) Which design of VQFlow do you think contributes to a better understanding of query requirements?

☐ Flow-based design (top-down reasoning).

☐ Modular requirements.

☐ Clear graphics.

☐ Others: _____.

7. (Multiple choice) What improvements do you expect in VQFlow?

☐ More easily use VQFlow.

☐ More clearly describe query requirements.

☐ Others: _____.

8. (Single option) Are you willing to learn to use VQFlow and recommend it to your colleagues?

○ Yes.

○ No.

9. (Q&A, optional) What additional learning costs do you think VQFlow has compared to the traditional method?

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10. (Q&A, optional) Do you have any additional comments or suggestions for VQFlow?

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